



Dr. Ernesto García Alfonso

Computational
Physicist/Chemist |
Molecular Modeling | HPC

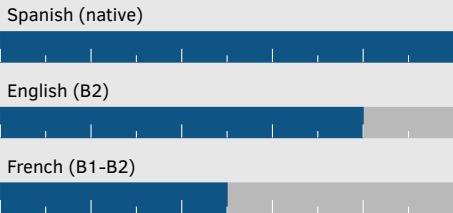
- November 14th, 1996
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- <https://github.com/ErnestoGarciaALfonso>
- <https://www.researchgate.net/profile/Ernesto-Garcia-Alfonso-2>

Technical skills —

- Programming: Python, C++, Fortran
- Simulation: MD (Tinker, GROMACS), DFT (Quantum Espresso), Helium Superfluid
- HPC: MPI, OpenMP, Linux
- Data: NumPy, analysis, workflows
- Tools: GitHub, LaTeX

About me —

Computational physicist/chemist with experience in DFT, molecular dynamics, and HPC (MPI-OpenMP). Skilled in scientific computing, with focus on simulation workflows and data analysis for complex molecular systems.



Professional Experience

Paris, France *Institut des Sciences Moléculaires d'Orsay – Paris-Saclay*
Present
Postdoctoral Researcher – Molecular Modeling

- Developed DFT and molecular dynamics workflows
- Performed electronic structure calculations
- Built Python pipelines for data analysis
- Executed simulations on HPC systems (MPI/OpenMP)
- Improved automation and reproducibility

Research Experience

- Atomistic simulations of molecular systems and clusters
- Modeling of electronic interactions in complex environments
- Structure–property relationships analysis
- International collaboration (France, Spain, Germany, Denmark, Cuba)

Education

2024–2025 : Conseil national de la recherche: CNR, Pisa, Italy Postdoc in Physics

- Quantum ESPRESSO for first-principles simulations, Density Functional Theory (DFT) for electronic structure calculations.
- ab-initio molecular dynamics to study atomic-scale dynamics.
- Free Energy Perturbation (FEP) methods for evaluating thermodynamic properties, and experienced in Nudged Elastic Band (NEB) techniques for exploring reaction pathways and transition states.
- GROMACS for classical molecular dynamics and the application of force fields in simulating molecular systems.

2021–2024 PhD in Physics – Toulouse, France

- Theoretical study of superfluid helium nanodroplet dynamics, focusing on cluster formation, ion solvation, Coulomb explosion, and quantum vortex nucleation and detection
- Superfluid $^4\text{He}_N$ nanodroplets, Helium Time-Dependent Density Functional Theory (He-TD-DFT)

2015–2020 Master in Physics – Havana University, Cuba

- Molecular dynamics and computational modeling
- C++ scientific programming

Publications

13 peer-reviewed publications in computational chemistry. <https://github.com/ErnestoGarciaALfonso/List-of-Papers/blob/main/Papers.pdf>

Grants and Awards

- 2025 – Scientific Work Award, Havana University
- 2024 – National Award, Cuban Academy of Sciences
- 2024 – E-COST Grant, NANO X Mobility Program
- 2023 – INSTEC Award

International Experience

- Research collaborations in France, Spain, Germany, Denmark, Cuba.
- Multidisciplinary international environments

Selected Conferences

- Oral presentations at international conferences
- Workshops on machine learning and molecular modeling